MYCOBACTERIAL WAG31 MEDIATES POLAR ORGANISATION AND CELL SHAPE BY INTERACTION WITH CONSERVED TRANSMEMBRANE PARTNERS

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Keywords: Wag31, Divisome, Membrane Protein Interaction, Structural Biology

ABSTRACT

The bacterial cell cycle requires a tight coordination of incorporation of new cell wall material at sites of elongation and division. While common model organisms like *Escherichia coli* and *Bacillus subtilis* elongate laterally, Mycobacteriales, including major human pathogens like *Mycobacterium tuberculosis* and *Corynebacterium diphteriae*, elongate from their cell poles. This polar growth is mediated by the elongasome machinery, organised around the essential conserved DivIVA-domain scaffolding protein Wag31, whose molecular mechanisms remain poorly understood despite more than two decades of research.

The membrane recruitment mechanism for DivIVA-domain proteins was thought to originate from a small peripheral membrane binding loop at the tip of the DivIVA dimer in a membrane curvature and lipid composition dependant manner (*B. subtilis* [1], *Clostridioides difficile* [2]). However, using *Corynebacterium glutamicum* as a model organism for *M. tuberculosis*, we showed that the DivIVA-domain of Wag31 is indispensable for proper localisation of Wag31, but surprisingly independent of membrane recognition [3].

Using Mass-spectrometry based interactomics and AlphaFold pull-down analyses, we identified two conserved transmembrane proteins, Wip1 and Wip2, that directly interact with Wag31. These proteins, which are essential in *Mycobacterium tuberculosis*, share two common and highly conserved motifs that mediate direct binding with the DivIVA-domain. Using

structural and *in vivo* approaches, we demonstrate that these proteins are required for proper Wag31 localisation, cell morphology, and normal cell cycle progression.

We present a revised model in which Wag31 is actively recruited to and retained near the polar and septal membrane through direct protein-protein interactions, independent of membrane curvature. In this model, the DivIVA-domain functions as a central hub for protein-protein interactions, coordinating cell pole organisation and shape maintenance.

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